IoT Challenge 2026-27

End-to-end system solution for the Internet of Things

Nature Match

IoT for Reconnecting with Nature

Members: Aimar Pagonabarraga, Alessandro Berti, Leire Larrauri

1. Description of the IoT system and its purpose

NatureMatch is a proposed IoT system that helps people choose outdoor places according to their intended activity, preferences, available time and personal access needs. Its central product is a connected monitoring station that measures visitor movements and environmental conditions. A web application uses the information generated by these stations to recommend suitable places and explain the reasons behind each suggestion.

The purpose is to reduce uncertainty about spending time in nature. Knowing where a park is does not necessarily tell someone whether its current conditions suit studying, running, relaxing or meeting friends. NatureMatch makes those conditions visible, helping students, staff and citizens incorporate nearby green spaces into everyday life.

Functionality

The system combines three functions. Firstly, it monitors the entry and exit of people to the spaces. Two infrared transmitter/receiver pairs establish successive detection points at an entrance, then the sequence of signal interruptions is processed to estimate whether a person enters or leaves. Starting from a known initial count, the system maintains an estimated occupancy. Secondly, it checks environmental conditions. Temperature and humidity, noise, light, UV and air-quality sensors provide information relevant to the outdoor activities the user can choose among. Finally, it allows for activity check-ins. An NFC reader allows visitors to register their intended activity voluntarily.

Regarding the entry and exit monitoring, it must be said that the infrared counter and NFC reader produce different information. The counter estimates total occupancy, including visitors who do not check in. NFC records declared activities among participating users. Public results would distinguish “estimated visitors” from “active voluntary activity registrations.” Check-out or session expiry would prevent registrations from remaining active indefinitely; an anonymous infrared exit cannot identify which activity session to close.

The environmental measurements would be interpreted according to the selected activity. Quiet studying would give greater importance to sound levels, brightness and comfortable conditions. Running would consider temperature, UV exposure and relevant air information. Someone seeking a social atmosphere could prefer a busier location. Occupancy and noise would remain separate measurements, since a busy place is not necessarily noisy.

Users could also specify essential requirements, such as step-free access, suitable path surfaces or the presence of rest facilities. These would be checked against surveyed place information before preferences are ranked. Bench and table presence would be recorded, but their individual availability would not be monitored in the MVP. The interface would support readable lists, screen readers and text alternatives to audio, but these will be add-ons, while monitoring of the environment and people counter remain the central figure in the system.

2. A typical real-life use case or scenario

Ane is a university student in Bilbao with a break between classes. She wants to study outdoors instead of remaining in the library. She prefers low noise and a less busy atmosphere, needs a step-free approach and wants a location where tables are present. She is uncertain about the current conditions of nearby green spaces.

Ane opens NatureMatch, selects “quiet studying” and enters her available time and maximum travel distance. The system first checks the recorded access characteristics and facilities. It excludes known conflicts with her essential requirements, then considers the available occupancy and environmental measurements.

At that point several monitored places are compared. In this case, given the situations and the corresponding requirements a campus garden might be suggested because its surveyed approach meets her requirements, its estimated occupancy is relatively low and its measured sound levels suit quiet studying. The recommendation would explain these reasons, display measurement times and clarify that individual table availability is unknown.

When Ane arrives, she can optionally select “studying” at the station and tap her NFC card. This creates a temporary activity registration. Other users can see the aggregate number of active studying check-ins without seeing her identity or personal access requirements. The infrared counter independently records her entry, so her registration is not added again to total occupancy.

If more people enter or local noise increases, NatureMatch updates its information. A user seeking quiet studying may receive a different assessment from someone looking for a social setting. UV information can also help people consider exposure when planning a longer stay outdoors. Having completed all the steps above, Ane saves any directions she needs and puts the application away. The intended change is that an outdoor study break becomes easier to plan and more realistic to repeat. The same process could support staff taking a short walk, students going running or citizens seeking company in a park.

3. Advantages over market alternatives or existing systems

In order not to be a simple environmental monitoring application NatureMatch could operate alongside established mapping and outdoor-discovery services. Its proposed value comes from combining information generated by its own IoT stations with activity-specific recommendations and personal access requirements, though this report focuses mainly on the IoT scope, due to its close relationship with the subject.

Google Maps already provides popular-time and live busyness information for some locations, using aggregated visit data. NatureMatch would generate entrance-based estimates for selected monitored green spaces and combine them with environmental measurements from relevant points. This could support locations chosen by a university or municipality without depending on sufficient participation in a location-history service.

AllTrails is another guides and routes platform which offers activity, difficulty, accessibility and visitation filters. Komoot supports sport-specific route planning, adjustable pace, surface and elevation information, and route-weather features. NatureMatch would focus on everyday participation in nearby nature, including studying and short breaks, while integrating measurements and voluntary activity registrations from its physical product.

Main advantages

The main proposed advantages of this solutions are the following:

- **A direct connection between sensing and recommendations.** Physical changes at the monitored location affect the information presented to users, making the IoT station fundamental to the service.
- **Different criteria for different activities and meaningful access requirements.** Studying, running, relaxing and socializing do not receive the same assessment. Users can prefer a quieter or busier setting according to their purpose. To that end they describe their practical needs which are then checked against recorded characteristics before preferences are ranked.
- **It could work as a complementary information source.** Entrance counting estimates use of the area, environmental sensors describe local conditions, and NFC provides voluntary activity information. None is presented as a substitute for the others.
- **Explained results and visible uncertainty.** Recommendations identify their reasons, measurement times and limitations. Activity registrations are distinguished from total occupancy, and experimental sensor readings are labeled accordingly.

Originality

It must be said that NatureMatch is primarily an innovation through integration. Infrared counting, environmental sensing, NFC communication, Raspberry Pi processing and web applications are established technologies. The project combines them around a specific need: helping people choose a nearby natural setting that suits their activity and requirements under current conditions.

Its originality lies in how the components work together. The station turns visitor movements, environmental changes and voluntary check-ins into actionable information. Users can make informed choices about where to study, exercise, relax or meet others, while institutions could use aggregate measurements to understand patterns of use. The MVP will test whether this combination produces useful, understandable recommendations with realistic course hardware.

4. Solution architecture diagram

The diagram shows the course MVP (solid boxes) and the scalable extension described in sections 6 and 7 (dashed box).

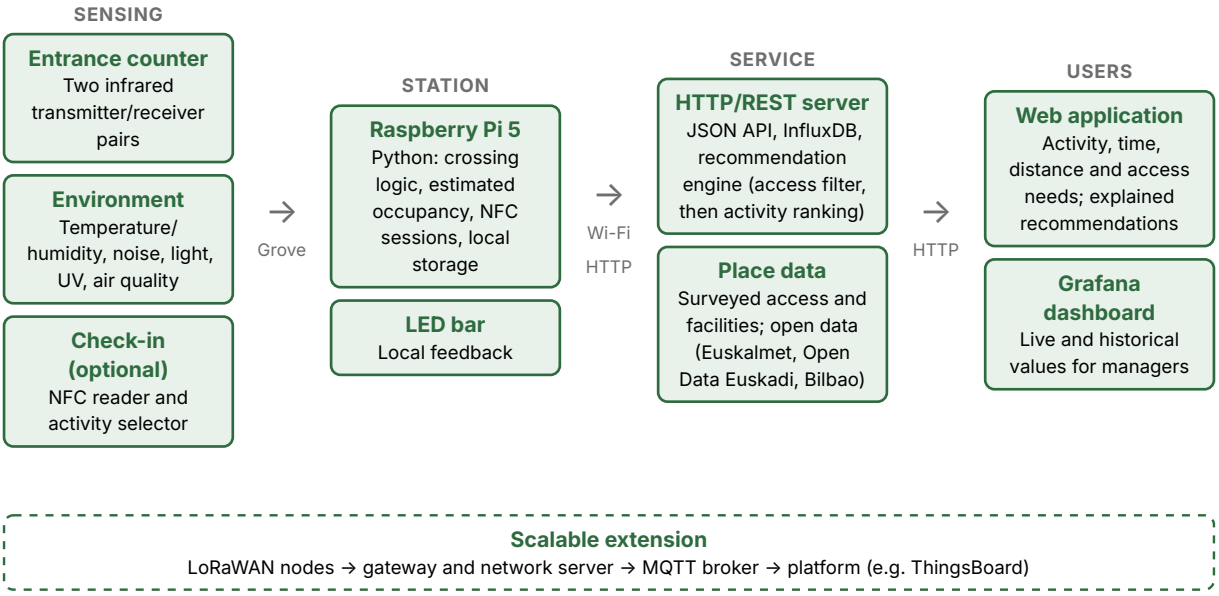


Figure 1: NatureMatch solution architecture.

5. Sensors and actuators

The broader vision is a network of NatureMatch stations across campus gardens, parks and other green spaces in Bilbao. Defined areas would require monitoring at every entrance to support occupancy estimates, as explained before. Environmental modules could be placed in the exact doors where the entrance monitoring is located or near relevant activity areas, while historical data could also help users understand how conditions change over time.

For the development of this solution the following Grove sensors will be used:

- One Raspberry Pi 5 and the appropriate Grove interfaces.
- Two infrared transmitter/receiver pairs.
- Temperature/humidity, noise, light, UV and air-quality sensors.
- One NFC reader with compatible test cards or tags and an activity selector.
- An LED bar for local feedback.

It should be noted that some of the sensors do not offer exactly the kind of information that would be useful for this system and that is the reason why the exact capabilities of the sensors would determine how their results are presented. The common Grove air-quality sensor provides a qualitative gas response rather than specific pollutant concentrations or an official Air Quality Index. The Grove UV sensor provides an approximate indicator that should be distinguished from an official UV Index. These experimental readings would be labeled clearly and could later be supplemented by authoritative environmental data.

Implementation could proceed in stages: entrance counting, environmental monitoring and recommendations, followed by optional NFC check-ins. Tests will compare crossing estimates with manual observations and examine pauses, reversals and different crossing speeds. Sensor reception, modulation, interfaces and voltage requirements will be checked before integration.

The way it has been described and designed, the MVP could demonstrate a complete sensing–analysis–feedback cycle: visitors entering or leaving, changing environmental readings and activity check-ins update the area's status, which changes the displayed information or recommendation. Measurements will include timestamps, and unavailable information will remain visible. Users can briefly consult the service, choose an outing and then focus on nature, fulfilling the main requirements of connecting with nature and then stepping aside.

6. Physical/link layer communication technologies

NatureMatch will use wired connections within each monitoring station and wireless communication between the station and the network. The course MVP will use the Raspberry Pi 5, the Grove interfaces used in class and Wi-Fi.

Connections within the station

Sensors and controls will connect through the appropriate digital, analog or I²C interfaces. Analog readings will pass through the Grove board's analog-to-digital converter before reaching the Raspberry Pi. The NFC reader will communicate with the Pi through its supported UART or I²C interface. Each module's voltage, connections and software compatibility will be checked before integration.

The infrared transmitter/receiver pairs will establish the optical detection points used for entrance counting. NFC will provide the short-range connection between a visitor's card or tag and the reader. These connections collect local information; the station's Wi-Fi connection carries information to other devices. Network connection for the MVP. The Raspberry Pi 5 supports dual-band Wi-Fi. It will connect to an approved campus network or a dedicated test access point, allowing users to reach its HTTP server from a browser. A local demonstration can operate without an Internet connection; Internet access will be needed for external data sources or a remotely hosted service. Coverage and device-to-device connectivity will be tested at the selected location rather than assumed. Raspberry Pi 5 specifications.

Other considerations

Since this solution will be implemented outdoors, there are other aspects to take into consideration as well. Regarding power, the MVP will use mains electricity and a suitable USB-C supply, such as the official Raspberry Pi 27 W supply. Actual consumption will be measured because supply capacity is not the same as energy use. Full monitoring stations will preferably remain mains-powered, supporting continuous entrance detection and NFC availability. Battery or solar operation would require a measured energy budget, sufficient autonomy and consideration of winter sunlight and shading.

Protection of the device against adverse weather conditions should also be bare in mind and, though it might not be strictly related to the implementation of the solution per se, initial outdoor tests will be supervised and sheltered. A permanent installation would require a suitable weather-resistant electronics enclosure, sealed cable entries, protected connectors and secure mounting. Sensor placement must preserve airflow, sound measurement, UV exposure and infrared alignment while protecting components from rain and condensation. The assembled installation would need testing.

Fianlly, when talking about coverage and expansion, Wi-Fi will remain suitable near buildings where reliable coverage exists. For suitable remote measuring nodes, the scalable design could use LoRaWAN, which supports low-power, wide-area communication. Those nodes would require additional radio hardware and a gateway with an Internet connection. Gateway coverage and installation locations would be checked on site. LoRaWAN is a future extension, outside the course MVP. Entrance events will be processed locally so counting can continue during a network interruption while the station remains powered. Accordingly, any battery-based design must preserve the continuous sensing needed for this function.

7. Application/data layer communication technologies

NatureMatch will use application-layer technologies to exchange measurements, recommendations and configuration requests between its monitoring stations, server and web interface. While Wi-Fi provides the network connection, these technologies define how the software communicates and how the information is organized.

Initial MVP implementation

For the MVP, we will use HTTP, the protocol used by web browsers to request information from servers. The Python application will collect and process the sensor readings on the Raspberry Pi 5, while the HTTP server developed in class will make the results available to the web interface. The browser will periodically request updated occupancy estimates, environmental measurements and activity-registration counts.

The server will provide a REST-style API, this is, an organized set of requests that other software can use to access its functions. For example, a GET request will retrieve current conditions or recommendations, while a POST request will submit a check-in, check-out or authorized settings change. NFC scans will be processed locally by the Raspberry Pi and recorded through the same session-management logic.

Information will be exchanged using JSON, a structured text format that both Python and browser applications can interpret. A measurement record will contain the station identifier, observation time, value, unit and quality status. This makes it possible to distinguish current measurements from outdated or unavailable information. Estimated occupancy and voluntary activity registrations will remain separate data fields.

Wider station implementation

For the wider implementation, we propose MQTT to connect multiple stations to the central service. MQTT uses a publish/subscribe model: stations publish messages under named channels, called topics, and a central broker distributes them to subscribed applications. For example, stations could publish environmental measurements and occupancy updates, while the monitoring dashboard subscribes to receive them. This supports several devices and services sharing information through one communication system.

MQTT would also support communication in the opposite direction. Stations could subscribe to configuration commands, allowing an authorized administrator to change reporting intervals or test an LED indicator. The device would validate the command and return an acknowledgement. The browser-facing application would continue using HTTP/REST, protected by HTTPS in deployment. Future LoRaWAN nodes would send their radio messages through a gateway and network server, whose backend integration would deliver the information through MQTT or HTTPS.

To handle connection failures, the station will continue processing entrance events locally while powered and store records for later synchronization. Timestamps and unique record identifiers will help prevent duplicate processing when messages are resent. The interface will show stale data explicitly, and any interruption that causes missed crossings will require reconciliation of the occupancy estimate.

Controlled classroom tests may use HTTP on an isolated local network. Deployment beyond that environment will use HTTPS and MQTT over TLS. TLS encrypts communication and, with certificate validation, helps verify the server's identity. Device credentials and authenticated administration will restrict who can submit data or change settings, protecting measurements, temporary check-in information and remote commands.

8. Cloud-based monitoring dashboard and data analytics

This section explains how NatureMatch would store, display and analyse the information collected by its monitoring stations. It describes the proposed dashboard for park managers, the analytics used to interpret occupancy and environmental conditions, and the open data sources that would complement the local measurements.

- **Dashboard MVP:** InfluxDB plus Grafana on a local server, showing live and historical values per station: estimated occupancy, environmental measurements and the recommendation score per activity. Scalable: ThingsBoard (device management, LoRaWAN integration, rules engine and dashboards) or Grafana Cloud, hosted in an EU region for data-residency reasons. Grafana is the "city view" for park management; end users shall only see the app.
- **Analytics:** (1) Recommendation engine: essential access requirements are checked first, then places are ranked with activity-specific weights. (2) Data-quality checks: out-of-range detection, stuck sensor and drift detection, and calibration of the noise sensor. (3) Trend analysis: typical occupancy and conditions of each space by hour and season, which feeds the forecast. (4) Validation: crossing estimates are compared with manual observations and thresholds are tuned accordingly; a simple approach is enough for the MVP. For the cloud, we start with a self-hosted container and consider a managed service (AWS IoT or Azure IoT) only if the number of stations justifies the cost.

Dashboard example

The example below illustrates a Grafana dashboard using station measurements and calculated activity scores stored in InfluxDB by the Python service. A manager could select a station and time range, inspect visitor trends and check sensor freshness. Estimated occupancy and voluntary NFC check-ins are shown separately; activity scores are illustrative rankings, not percentages of visitors. All

displayed values are fictional demonstration data, and the dashboard is a design proposal rather than an implemented system.

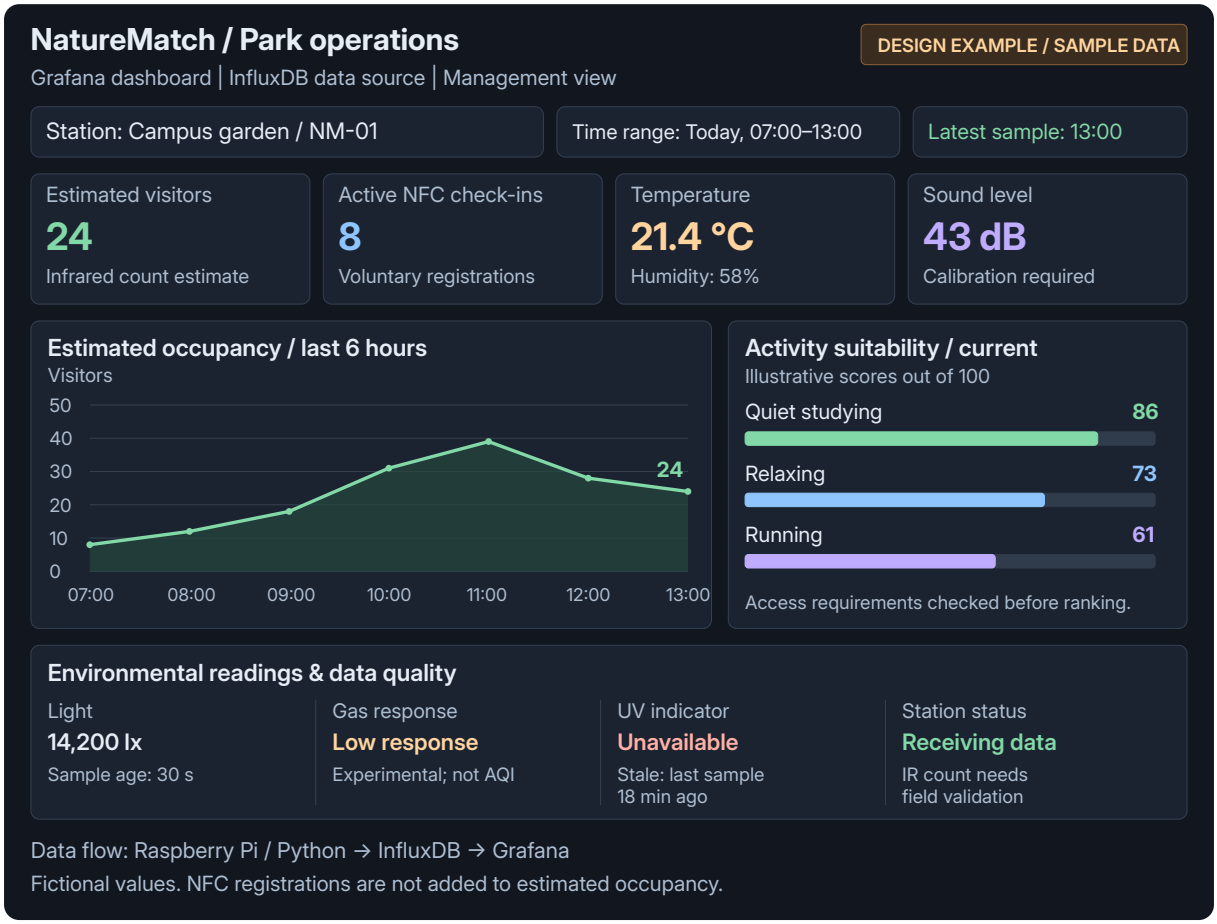


Figure 2: Proposed NatureMatch management dashboard using Grafana and InfluxDB. AI-generated illustration (OpenAI, 2026). Fictional data; gas and UV readings remain experimental indicators.

Open data sources

- **Euskalmet:** weather forecast, rainfall, wind and UV forecast, used to predict conditions for the next hours.
- **Open data Euskadi:** air-quality station data, used as a reference to validate and calibrate our gas sensor.
- **Bilbao open data portal:** location, size and facilities of parks and green spaces; accessibility and surface information used to check users' essential access requirements.

9. Ethics, privacy and environmental impact

- **Privacy:** The system is privacy-by-design. Privacy is a sensitive topic when monitoring people in public spaces, which is why there will be no cameras or microphones. The noise sensor computes a level in dB and never stores or transmits audio. The infrared counter only registers the sequence of beam interruptions at an entrance, which cannot identify anyone. NFC check-ins are voluntary and use temporary session identifiers, and personal access requirements are never shown to other

users. The app works without an account; location is used on the phone to filter nearby spaces and it's not stored. Any optional account or notification data follows GDPR (consent, minimization, deletion on request).

- **Environmental footprint:** The MVP stations are mains-powered with a measured energy budget; any future battery or solar variant would use low duty cycles. Stations use a modular and repairable design, and are collected under WEEE rules at end of life. We limit the number of stations by choosing representative locations and reusing existing posts. Stations are placed away from nesting areas, and the LED bar is dim and switched off at night to avoid disturbing wildlife and to save energy.
- **Screen use:** The service is designed to be consulted briefly: users choose an outing and then focus on nature, and the LED bar gives local feedback at the station without a phone.

10. Business model

Canvas block	Proposed solution
Customer segments	Students and staff (primary, campus) Citizens: runners, families, people with reduced mobility, newcomers and tourists City parks and environment departments (B2G customer and data user)
Value propositions	Citizens: a green space that suits my activity and access needs, right now City: continuous, objective data on use and conditions of green spaces; spreads visitors to under-used areas University: a wellbeing service that gets people outdoors
Channels	Web application with no installation QR code and LED bar at the stations University and city communication channels
Customer relationships	Free app, no account needed Voluntary NFC activity check-ins at the station Co-design with schools, sports clubs and accessibility associations
Revenue streams	Annual subscription from city or university for the data service and dashboard Non-intrusive sponsorship by sports and health organisations Public funding for green infrastructure and sustainability projects
Key resources	Monitoring stations and gateways Own dataset plus open data and surveyed place information Software team and recommendation models
Key activities	Station deployment and maintenance Data platform and recommendation engine App development and sensor calibration
Key partners	Ayuntamiento de Bilbao (parks and environment) The university LoRaWAN community networks Hardware and enclosure suppliers
Cost structure	Hardware and installation Connectivity and cloud hosting Maintenance (cleaning, calibration, replacements) and development

Table 1: Business Model Canvas.

11. Initial task planning

We work in short sprints with a working increment at the end of each, following the implementation stages described in section 5. Roles: Product Owner (rotating, owns the value and the backlog), Scrum Master (runs the sprint events) and developers (all members). Tasks are tracked on a shared board and reviewed weekly with the teacher.

Sprint	MVP	Goal and main backlog items
1 (weeks 1–2)	MVP 0: entrance counting	Wire the two infrared pairs to the Raspberry Pi 5; detect crossing sequences and estimate occupancy; compare with manual counts
2 (weeks 3–4)	MVP 1: environment	Read temperature/humidity, noise, light, UV and air-quality sensors in Python; add timestamps and local logging; label experimental readings
3 (weeks 5–6)	MVP 2: data flow	HTTP/REST server with GET and POST (JSON) and storage; InfluxDB and Grafana dashboard
4 (weeks 7–8)	MVP 3: recommendation	Activity profiles and access-requirement filter; web application with explained recommendations; LED bar feedback
5 (weeks 9–10)	Final MVP	Optional NFC check-ins and sessions; field test at one entrance; documentation and final presentation

Table 2: Sprint plan.

Main risks

Among the most remarkable risks that should be considered are: counting accuracy (mitigated by tests against manual observations), sensor limitations (mitigated by calibration and clearly labelled experimental readings), outdoor weather exposure during tests (short, supervised tests, simple enclosure) and scope creep (NFC check-ins are optional and the Product Owner freezes the backlog after sprint 3).

12. References and links

- Basque Government (n.d.) *Open Data Euskadi*. Available at: https://opendata.euskadi.eus (Accessed: 15 October 2026).
- Bilbao City Council (n.d.) *Bilbao Open Data*. Available at: https://www.bilbao.eus/opendata (Accessed: 15 October 2026).
- European Parliament and Council of the European Union (2016) 'Regulation (EU) 2016/679 (General Data Protection Regulation)', *Official Journal of the European Union*, L 119, 4 May, pp. 1–88. Available at: https://eur-lex.europa.eu/eli/reg/2016/679/oj/spa (Accessed: 15 October 2026).
- Euskalmet (n.d.) *Euskalmet: Euskal meteorologia agentzia*. Available at: https://www.euskalmet.euskadi.eus (Accessed: 15 October 2026).
- Grafana Labs (n.d.) *Technical documentation*. Available at: https://grafana.com/docs/ (Accessed: 15 October 2026).
- OpenAI. (2026, October 15). *Document edition with Typst* [Generative AI chat]. Codex.
- Schwaber, K. and Sutherland, J. (2020) *The Scrum Guide*. Available at: https://scrumguides.org (Accessed: 15 October 2026).
- Seeed Studio (2022) *Grove - NFC*. Available at: https://wiki.seeedstudio.com/Grove_NFC/ (Accessed: 15 October 2026).
- Seeed Studio (2025) *Grove Base Hat for Raspberry Pi*. Available at: https://wiki.seeedstudio.com/Grove_Base_Hat_for_Raspberry_Pi/ (Accessed: 15 October 2026).
- The Things Network (n.d.) *Learn*. Available at: https://www.thethingsnetwork.org/docs/ (Accessed: 15 October 2026).
- ThingsBoard (n.d.-a) *RPC - MQTT API*. Available at: https://thingsboard.io/docs/reference/mqtt-api/rpc/ (Accessed: 15 October 2026).
- ThingsBoard (n.d.-b) *Telemetry - MQTT API*. Available at: https://thingsboard.io/docs/reference/mqtt-api/telemetry/ (Accessed: 15 October 2026).
- ThingsBoard (n.d.-c) *ThingsBoard CE Docs*. Available at: https://thingsboard.io/docs/ (Accessed: 15 October 2026).